



Bonds and boundaries: Network structure, organizational boundaries, and job performance

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ABSTRACT

We propose and test a framework that describes the relationship between network structures and job performance. We provide an integration of the current conceptualizations of social capital as they pertain to job performance outcomes by taking a multi-dimensional view of job performance. We break down job performance into creativity, decision-making, task execution, and teamwork, and distinguish the effect of structural holes within and across the organizational boundary on these four job performance domains. In an analysis of 318 managers, we find that networks rich in structural holes that cross the organizational boundary had a positive association with creativity and decision-making, whereas networks with few structural holes within the organization had a positive association with task execution and teamwork. We discuss the theoretical implications for integrating the social capital, boundary spanning, and network structure literatures, as well as the practical benefits of giving much more precise advice to managers and employees regarding how to use networks to improve performance at work.

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Introduction

How are managers' professional networks related to their job performance? The extant literature might be characterized as suggesting two "good options" for network structures that affect job performance (Gargiulo & Benassi, 2000). One common conclusion has been that it is good to connect others who are not themselves connected (i.e., to span structural holes), as this leads to improved job performance through the mechanisms of accessing valuable information (Brass, 1984; Ibarra, 1993) and generating novel ideas (Burt, 2004). On the other hand, there are theoretical arguments for why the absence of structural holes (i.e. network closure) may help job performance through the mechanism of increased social cohesion (Coleman, 1988), and evidence to support these ideas (e.g., Baker, 1984; Gargiulo, Ertug, & Galunic, 2009). In sum, while structural holes provide superior access to information and greater opportunities to exercise control (e.g. Burt, 1992, 1997), network closure facilitates cooperation efficiency through repeated interactions and dense interconnection (e.g., Coleman, 1988; Obstfeld, 2005). Given that any particular triad must be either open or closed, managers might understandably wonder just how to structure their networks to achieve better job performance. We offer an answer to that question based on two contingencies: (1) what form

of job performance is desired; and (2) where the relevant network relationships are relative to the organizational boundary.

The idea that structure holes and closed networks both do good, but different, things for job performance has achieved taken-for-granted status in the management literature. This is largely due to a set of influential studies that have shown contingencies regarding the impact of structural holes. For example, both structural holes and closed networks help job performance depending on, respectively, the form of utility the networker recognizes in the relationship (Podolny & Baron, 1997), whether they are among current or old ties (Soda, Usai, & Zaheer, 2004), and whether they are embedded in individualistic or collectivist cultures (Xiao & Tsui, 2007). In each case, the fundamental theoretical mechanisms espoused are the same, that structural holes provide divergent information while network closure facilitates cooperation efficiency. However, empirically these studies look at overall performance outcomes, and do not directly document the espoused mechanisms of divergent information and cooperative efficiency. The approach of using overall job performance in network studies is common, as when job performance is measured by summary indexes such as in-role (Cross & Cummings, 2004; Mehra, Kilduff, & Brass, 2001; Papa, 1990; Roberts & O'Reilly, 1979; Rodan & Galunic, 2004) vs. extra-role performance (Settoon & Mossholder, 2002; Sparrowe, Liden, Wayne, & Kraimer, 2001), or by examining outcomes that depend on overall performance, such as salary increases, promotions, bonuses and career advancement (Burt, 1997; Gargiulo et al., 2009; Mizuchi, Stearns, & Fleischer, 2011;

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Seibert, Kraimer, & Liden, 2001). All of these studies have the advantage of focusing on dependent measures that clearly matter for networkers, but in all of them, the dependence on an overall job performance measure means that evidence as to the specific mechanisms through which networks influence performance must be indirect.

There are other studies examining a sub-dimension of job performance rather than overall performance. Usually these have been sub-dimensions that depend on information diversity such as creativity (e.g. Burt, 2004; Hargadon, 2002; Hargadon & Sutton, 1997) or flexibility in the face of organizational change (Gargiulo & Benassi, 2000) rather than on coordination efficiency (but see Obstfeld, 2005). Empirical studies such as these that examine one sub-dimension of job performance are better able to expose the mechanisms through which structure holes or closed networks operate, but they cannot show whether and how individual networkers can simultaneously enjoy the divergent information benefits from structural holes and the coordination efficiency from network closure. To address both the need for direct evidence of mechanisms, and evidence as to whether and how structural holes and closure can operate in the same network, we disaggregate job performance into dimensions that benefit from divergent information (i.e., creativity, decision-making), and others that benefit from coordination efficiency (i.e., task execution, teamwork). By directly examining the effects of structural holes and network closure on different outcomes, we seek to theoretically and empirically locate the impact of structural holes and closed networks on specific dimensions of job performance.

Job performance is also a particularly interesting outcome measure because performance evaluations are local, in that employees are compared to others in the same organization, rather than solely to some objective standard or personal achievement (Salanick & Pfeffer, 1978; Viswesvaran, Ones, & Schmidt, 1996). This means that structural holes that enable an employee to be the unique person to bring fresh information to the organization may be most valuable for differentiating that employee from his or her peers. For example, information available to ego from a colleague at work or from someone outside the organization that the colleague also knows may lead to less competitive advantage to ego, because he or she cannot uniquely represent the information in the organization. Alternatively, it may not matter for job performance evaluation whether ego's contacts outside the organization know each other, because ego does not compete with them inside the organization. In other words, whether ego and his/her alters are in the same or different social groups (in this case, the organization) changes the functionality of a structural hole (see also, Gould & Fernandez, 1989). We thus emphasize the interplay of structural holes and the organizational boundary, differentiating structural holes inside the organization from those across the organizational boundary¹ (see Fig. 1).

We propose that the effect of structural holes on each job performance dimension depends on whether the holes are inside or cross the organizational boundary. Creativity and decision-making are enhanced by divergent information that could potentially be accessed from structural holes inside or outside the organization. However, the benefit of structural holes for creativity and decision-making depends on heterogeneity and non-redundant knowledge across the holes (Mors, 2010; Rodan & Galunic, 2004), and these conditions are more likely to be found when the hole spans the organizational boundary than when it is within the organization. On the other hand, teamwork and execution occur overwhelmingly within organizations, and we argue that they are

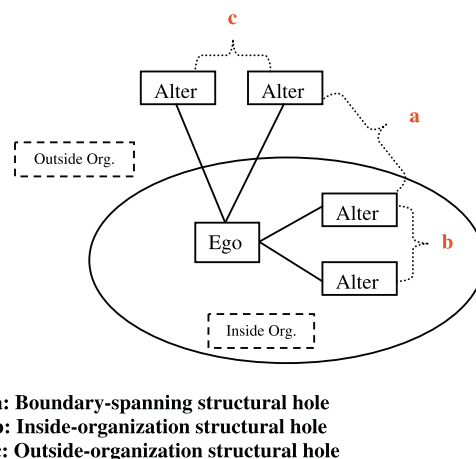


Fig. 1. Three types of the structural holes.

positively influenced by closure among contacts in the same organization, and not at all by contacts outside the organization. In sum, our key contribution is to articulate and examine two contingencies of the link between network structure and job performance: (1) What dimension of job performance is being explained, and (2) where is the structural hole—inside, outside, or across the organizational boundary? As a result of considering these two contingencies, we are able to generate theoretical predictions as to just what structures, in what parts of a network, are associated what types of job performance.²

Structural holes spanning organizational boundaries

The structural hole at the boundary secures the advantages that stem from having unique contacts outside the organization. Having a larger number of ties across organizational boundaries is associated with the adoption of less normative strategies (Geletkanycz & Hambrick, 1997), the acquisition of competitive capabilities (McEvily & Zaheer, 1999), and product innovation (Hargadon & Sutton, 1997; Reagans & Zuckerman, 2001). In terms of predicting job performance, Cross and Cummings (2004) found that having ties outside the organization contributes to employees' overall performance, above and beyond the effect of centrality in the network. Likewise, in a sample of research scientists, Perry Smith (2006) found that the number of outside ties could compensate for the disadvantages of having low centrality at work. Scientists with low network centrality at work but with a large number of outside ties achieved equally high creativity performance as those who have high centrality.

Nevertheless, the advantages of outside connections may not necessarily become performance advantages. Consider a case for a lunch get-together where ego has invited one colleague and one friend outside of work who did not know each other previously. The meeting of a colleague and a friend would not change the number of ego's contacts outside of work, but such a meeting would close the structural hole at the organizational boundary, assuming that a tie is established between the colleague and the friend. This closing of a structural hole could impose significant job performance constraints on ego, because any information and opportunities that ego could obtain from the friend are then more

¹ In this paper we treat the connectedness of extra-organizational ties as a control, rather than the basis of theory.

² Table 1 presents a summary of our arguments regarding the impact of boundary-spanning and inside-organization structural holes (and the outside organization structural holes which are controlled in our analysis) on the dimensions of job performance that depend on accessing divergent information and on coordination efficiency.

Table 1
Summary of the distinct functionality of structural holes.

	Benefit from access to divergent information	Cost of lost coordinative efficiency at work
Boundary (structural holes at position "a")	High: Ego is the sole source of unique and non-redundant information to the organization (<i>Hypotheses 1 and 2</i>)	Extraneous: Alters outside the work organization is less likely to be involved in routine tasks at work (<i>control variable</i>)
Inside the organization (structural holes at position "b")	Medium: The unique information from a colleague is still redundant to the organization (<i>control variable</i>)	High: Disconnected alters at work are less conducive to mobilizing trust and support within the organization (<i>Hypotheses 3 and 4</i>)
Outside the organization (structural holes at position "c")	High: Ego that is connected to distinct social groups is likely to have more heterogenous information (<i>control variable</i>)	Extraneous: Alters outside the work organization is less likely to be involved in routine tasks at work (<i>control variable</i>)

likely to be available to the colleague as well. So it is not only the number of cross-boundary connections that matters, but also the degree of disconnection between the contacts inside and outside the organization. If outside contacts are also connected with colleagues at work, the information and opportunities they offer are less likely to be uniquely available to ego. Furthermore, the benefits that come from connecting distinct social circles inside and outside the organization may not be uniformly conducive to promoting all kinds of performance at work. Specifically, we hypothesize that structural holes across the organizational boundary are likely to lead to better performance in terms of creativity and decision-making.

Creativity

The disconnections between the circles inside and outside the organization that constitute structural holes bring opportunities for distinct experiences in different social spaces. Connecting such distinct groups enables an employee to get access to a greater number of novel ideas, different perspectives and more diverse information. Individuals can be creative simply by bringing new information or novel concepts from contacts outside the organization to their work group colleagues. For example, Burt (2004) has provided evidence from the networks around managers in a large American electronics company showing that between-group brokers are more likely to come up with new ideas that are evaluated positively. Although some work has pointed out that open networks are not perfect surrogates for more knowledge heterogeneity (Mors, 2010; Rodan & Galunic, 2004), diverse knowledge is generally more likely to arise from connections outside than inside the in-group.

Furthermore, on a cognitive level, the distinct social circles that create structural holes at an organization's boundary may yield benefits resembling dual identity effects. For example, superior creativity is relatively prevalent among first- and second-generation immigrants (Lambert, Tucker, & d'Anglejan, 1973; Simonton, 1997), among bilingual people (Nemeth & Kwan, 1987; Simonton, 1999), and among people with multiple cultural identities (Leung, Maddux, Galinsky, & Chiu, 2008; Maddux & Galinsky, 2009). This is partly because distinct experiences from disconnected social circles can increase an individual's psychological readiness to recruit and accept ideas from unfamiliar sources, thus facilitating information recombination (Scholer & Melcher, 1995) and conceptual expansion (Ward, Smith, & Finke, 1999), both of which are central to the creative process. Therefore, from the perspectives on information diversity and cognitive processing, managers who bridge contacts inside and outside the organization should, on average, have better creativity performance.

Hypothesis 1. In a professional network, a high degree of structural holes across the organizational boundary is positively associated with better performance in creativity.

Decision-making

Individuals are not isolated decision-makers but rather are influenced by their network of relationships. As with creativity, structural holes at the organizational boundary create opportunities for the manager to better contribute to the decision-making process by bringing information from outside the organization to bear on workplace problems. Additionally, bridging structural holes at the boundary also allows an employee to exercise control in integrating perspectives from distinct social circles. This is likely to put employees who uniquely connect contacts inside and outside the organization in a position with benefits resembling those of integrative complexity. Integrative complexity has two components: differentiation, which refers to recognizing multiple perspectives about an issue; and integration, which refers to understanding the conceptual relationships among those perspectives (Suedfeld, Tetlock, & Streufert, 1992). Arguably, the more variegated the contacts one is exposed to, the greater the potential for differentiation and integration. Decision-makers comfortable with high integrative complexity are less likely to fall prey to cognitive biases and are less likely to be trapped by group thinking (e.g., Antonio et al., 2004; Tetlock, 1985). Related, Gargiulo and Benassi (2000) found that managers who had less constrained networks made better decisions when adapting to new challenges. On average, managers who bridge more structural holes at their organizations' boundaries should show better decision-making performance.

Hypothesis 2. In a professional network, a high degree of structural holes across the organizational boundary is associated with better performance in decision-making.

Within-organization connections

Prior research on individuals' social networks within the organization has focused on idea generation and information access to argue that employees with many and unique intra-organizational contacts have advantages in problem solving (Baldwin, Bedell, & Johnson, 1997; Mehra et al., 2001; Sparrowe et al., 2001). Managers have been shown to derive power and influence from reaching out to a wide range of organization members (Brass, 1984; Ibarra, 1993; Ibarra & Andrews, 1993). Structural holes within the organization have also been linked with early promotion and higher salary (Burt, 1992). But such analysis does not consider the low level of cohesion associated with structural holes, which can entail costs for getting things done within an organization (Gargiulo et al., 2009). In this regard, the closure of structural holes can improve communication quality (Baker, 1984) as well as make it less risky for people in the network to trust one another (Coleman, 1988). These are critical factors for effective task performance and team work. We argue that managers spanning fewer structural holes within the organization should in general achieve better task execution and teamwork.

Task execution

Managers whose contacts within the organization are themselves highly interconnected have several advantages in task execution, such as following through with schedules and staying focused on tasks. Specifically, a high degree of interconnection enhances interdependence and reduces the need for monitoring. In a cohesive social network, individuals are less likely to behave opportunistically because information about their behavior is readily accessible to others (Chua, Ingram, & Morris, 2008; Granovetter, 1985). Interconnected others protect ego from exploitation because they can act in concert against those who violate their group's norms of conduct. A cohesive network also facilitates building a strong culture that shapes how all members perceive problems, interact, approach decisions, and solve problems, as well as motivates individuals to forgo self-interest when it is in conflict with the organization's goals (O'Reilly & Chatman, 1996; Trice & Beyer, 1993). Furthermore, interconnections within an organization contribute to the construction of a clear identity and cohesive role expectations. An identity, building on an interconnected network, provides a coherent language and discourse, consistent interpretation of symbols and coordination in daily work routines. It is useful not only for supporting coordination, but also for maintaining a dialogue through which information and solutions can be discovered (Dutton, Dukerich, & Harquail, 1994).

Getting things done requires managers to communicate and share information effectively. Ample research has shown that structural holes act as barriers to knowledge transfer (e.g. Fleming & Waguespack, 2007; Hansen, 1999). For example, Tortoriello and Krackhardt (2010) analyzed scientists and engineers involved in research and development work and revealed that those who bridged unit boundaries were more likely to generate innovations when they shared common third-party ties. In such a situation a cohesive network is particularly relevant in facilitating information sharing and collaboration within an organization. Likewise, one challenge in sharing information is the information-sampling bias—the probability that a given piece of information will be shared among a group decreases as the number of members who are aware of it decreases (Stasser & Stewart, 1992). Clark and Kashima (2007) have shown that one critical cause of information-sampling bias is that people focus on building social bonds through communicating common knowledge and are averse to speaking about unique information. People in a highly connected group tend to have stronger social bonding and relatively more stable relationships (Krackhardt, 1998), which may diminish the need to focus on common knowledge and thereby reduce the information-sampling bias. This suggests that employees embedded in a cohesive network at work should be better able to pool a larger set of information to get tasks done. All of these considerations should make managers embedded in cohesive networks with fewer structural holes at work better able to get things done.

Hypothesis 3. In a professional network, a low degree of structural holes within the organization is associated with better performance in task execution.

Teamwork

Similar logic applies to teamwork. Regardless of the job level and complexity, organizational members have to collaborate with colleagues and are often involved in one or more teams simultaneously. Such behaviors in working with team members are a critical component in various job performance measures (Borman & Motowidlo, 1997; Campbell, 1990; Welbourne, Johnson, & Erez, 1998). While prior research has focused on optimal network

structure for high performance at the level of the team (e.g., Balkundi & Harrison, 2006; Oh, Chung, & Labianca, 2004), it is also important to look at networks and individual teamwork behaviors that collectively constitute team performance. First, people who are embedded in a cohesive network in the workplace are more likely to elicit greater reciprocity and trust (Granovetter, 1985; Krackhardt, 1990; Portes & Sensenbrenner, 1993). Such an environment not only encourages cooperative behavior, but also sanctions against free-riding and self-serving behavior (Coleman, 1988; Levine, 1991). It encourages risk taking in offering help to teammates, because people know such favors will be highly regarded and are more likely to be returned (e.g., Edmondson, 1999; Kramer, Hanna, Su, & Wei, 2001).

A cohesive network in the workplace that goes beyond the scope of the team is more likely to moderate some of the costs that a highly closed team network could impose. A strong ingroup connection tends to foster ingroup favoritism and negative outgroup biases, which can limit the group's effectiveness (Pruitt & Rubin, 1986; Tajfel & Turner, 1985). Cohesive connections with others outside the team facilitate communication with other outgroup members within the firm, which can be critical for getting buy-in for the team (Oh, Labianca, & Chung, 2006). A cohesive network within the organization also enables a manager to monitor teamwork more effectively through a broader range of interconnected contacts. Previous studies have usually focused on the aggregate performance of teams, but here we are addressing a positive influence on the perceived teamwork performance of individuals. We expect to see such an influence on individual teamwork performance among those who are embedded in a cohesive intra-organizational network.

Hypothesis 4. In a professional network, the degree of structural hole within the organization is negatively associated with an individual's teamwork performance.

Method

Participants and procedure

We collected the data from managers attending an Executive-MBA program at a business school in a large city in the United States. The students were managers who continued fulltime work while they studied. A total of 335 managers participated in this study, and 318 of them finished all the surveys. We excluded the observations with missing data from the subsequent analysis. Of the 318 managers, 28% were female, 55% were Caucasians, 38% were Asians (Chinese, Japanese, Korean or Indian), and the rest were of other races (mostly African-Americans and Hispanic Americans). Their most common industries of employment were finance and banking (42%). Typically, the participants held managerial positions in banks, other financial institutions, or other professional service firms. Some participants held executive positions in smaller companies (e.g., as CEO of a family business). A smaller group were professionals who had risen to supervisory or managerial positions (e.g., a PhD scientist who led a research project for a large pharmaceutical company). At the beginning of their MBA program, the participants were instructed to request job performance feedback from their work colleagues as part of a 360° feedback exercise. Participants nominated at least four colleagues from work and provided their email addresses. A private link for a web survey was then sent directly to the colleagues. The survey was completed on-line and anonymously. Participants received an aggregated performance rating from all the raters, who were not individually identified. At approximately the same time, the participants completed a network survey as part of a separate exercise. Each participant was given feedback on his or her network profile.

Performance measures

As part of the 360-survey, participants were rated by work colleagues on their creativity, decision-making, task execution, and teamwork. Items for measuring creativity were from the creativity dimension under the generalized role-based performance scale (Welbourne et al., 1998). Items for the other dimensions were developed with the goal being to allow for comparisons across jobs and across companies (Austin & Villanova, 1992). On average, participants had eight work colleagues serve as raters. We asked raters to clarify how well they knew the participant, using a four-point scale, ranging from “not well at all” (1) to “extremely well” (4). The average score on the familiarity measure was 3.02 ($SD = 0.36$). The colleagues rated the participant's performance with three items for each of creativity, decision making, execution and teamwork, using seven-point scales ranging from “not at all” (1) to “very much” (7). Appendix A lists all items. The Cronbach alphas were 0.85 for creativity; 0.74 for decision-making; 0.79 for execution, and 0.67 for teamwork.³

Because each participant's performance was rated by more than one rater, intra-class correlation coefficients ($ICC(1)$), within-group inter-rater agreement (multi-item $R_{WG(J)}$; James, Demaree, & Wolf, 1993), and indices of average deviation from the scale mean ($AD_{M(J)}$; Burke, Finkelstein, & Dusig, 1999) were used to assess whether or not the judges were in sufficient agreement in their ratings to justify aggregation of their responses. As indicated in Table 2, the raters were sufficiently consistent to justify aggregation in all cases. Specifically, LeBreton and Senter (2008) recommend that the traditional conventions used when interpreting effect sizes also be used when interpreting $ICC(1)$ values: a value of 0.01 is considered a “small” effect, a value of 0.10 is considered a “medium” effect, and a value of 0.25 is considered “large.” The $ICC(1)$ values of these raters ranged from 0.19 to 0.38, which represent medium to large effects. An $R_{WG(J)}$ value of 0.7 or above is considered necessary to demonstrate within-group agreement ($R_{WG(J)} = 1.0$ indicates perfect agreement). All of these variables exceeded this cutoff. An $AD_{M(J)}$ value of 1.2 is considered the upper limit for acceptable inter-rater agreement with a seven-point Likert-type scale (Burke & Dunlap, 2002). The $AD_{M(J)}$ index measures the dispersion of responses above the mean; therefore, a smaller score indicates better agreement. All variables met this criterion.

To examine the discriminant validity, we conducted a confirmatory factor analysis. Using LISREL 8.54 (Jöreskog & Sörbom, 1993), we developed three models associated with four performance constructs and represented the constructs as latent variables with items as indicators of each latent variable. The most restrictive model (the null model) consisted of 12 orthogonal factors that corresponded to the 12-item scale. The second model allowed the 12 measured items to load on one factor. The third model allowed the first six items measuring creativity and decision-making to load on one factor, whereas the second factor included the last six items measuring execution and teamwork. In the fourth model, which featured our four performance dimensions of creativity, decision-making, execution, and teamwork, the three items associated with their respective factors were allowed to load on those factors. If the four factors are distinct, the fourth model should have the best fit. Table 3 reports these results. We compared the fit among the models using chi-squared tests (Hayduk, 1987; Yuan & Bentler, 2004). As shown in Table 3, compared to the null, the one-factor, and

Table 2

Interrater agreement indices.

Variable	$ICC(1)$	$R_{WG(J)}$	$AD_{M(J)}$
1. Creativity	0.24	0.97	0.71
2. Decision-making	0.19	0.96	0.43
3. Task execution	0.23	0.96	0.61
4. Teamwork	0.38	0.97	0.58

Table 3

Fit Indices for nested measurement models.

Model	χ^2	df	CFI	RMSEA
Null model	2967.95***	78	0.42	0.324
One-factor model	577.27***	54	0.85	0.15
Two-factor model	344.46***	53	0.90	0.12
Four-factor model	202.29***	48	0.94	0.055

the two-factor models, the four-factor model was significantly better. This model also passed the two essential cutoff criteria for the model fitness test: the CFI (Bentler, 1990) is bigger than .9, and the root mean square error of approximation (RMSEA) does not exceed .055 (Steiger & Lind, 1980).

Network measures

The network survey allowed each participant (ego) to list up to 24 contacts (alters) whom they deemed most important in their professional network. They were instructed to list contacts who were significant in their professional networks in one or more of the following roles: providing economic resources, task information, career advice, and social support. Furthermore, alters could come from any context and were not restricted to the workplace. For each alter listed, the participant was asked to furnish details on the alter's background and the nature of their relationship. On average, each participant listed 22 alters ($SD = 3.83$). Participants were also asked how many more alters they would have listed if they had not been limited to 24, and responded on average, 28 more. Our process of generating network contacts is relatively unconstrained compared to others that ask respondents to list some alters in each of a few categories, such as “who do you go to for career advice” and “who do you go to for task advice” (e.g., Burt, 1992; Podolny & Baron, 1997). We used this unconstrained approach because we are interested in multiple dimensions of job performance, across a range of jobs, and so wanted to allow respondents the flexibility to report as many (up to 24) contacts that fulfilled whatever functions the respondent viewed as most important. Past papers that use an unconstrained name-generator, as we do, include Seibert et al. (2001).

Alter's structural position

We asked whether each alter was (a) within ego's work unit, (b) not in ego's work unit but within ego's organization, or (c) outside ego's organization. The subsequent analysis focused on contacts within or outside the organization and did not differentiate between (a) and (b) responses. A supplementary analysis differentiating (a) and (b) did not yield significantly different results from those reported here. We discuss the null effect of differentiating the work unit boundary in our sample later in the ‘General discussion’ section.

Relationship among alters

After participants had completed the first part of the survey which involved listing the alters and describing their relationships, the second part of the survey began with the heading, “Who knows whom in your network?” Here, the respondents filled in a

³ Although the Cronbach alpha for the teamwork dimension is below 0.7, we believe that it is partially because this measurement is relative sensitive to the number of items. In this study, the average correlation among the three items is 0.435. Based on the cronbach alpha formulate outlined by Cortina (1993), this average correlation would lead to an alpha score around 0.7 if there were 4 items and 0.8 if there were 6 items.

half-matrix indicating any positive relationships between alters. Negative relations were also indicated, but these were rare, comprising approximately 1% of the alter pairs, and we do not analyze them here. The survey instructions stated that, “A positive relationship can be (a) a close relationship (example: when people work very close together or have a high level of friendship) or (b) a positive but not especially close relationship (example: people who know each other but are not in frequent contact, and are not strong friends or enemies).”

Independent variables

We used Burt's (1992) measure of network constraint to capture the network structure.⁴ The constraint measure reflects a network rich in structural holes when the resulting value is low, and reflects network cohesion when it is high. It is correlated with network density, but also considers the relative strength of ego's ties to alters and the alters' ties to each other, as well as the degree to which certain alters play more central roles in the alter-alter network.

Constraint has two components. The first is the proportion of ego's (i 's) total relational strength that i directly allocates to j (an alter). The second component is the strength of the indirect connection between i and j through mutual contacts q (another alter), where p_{iq} is the proportion of his/her total relational strength that i devotes to q , and p_{qj} is the proportion of his/her total relational strength that contact q devotes to contact j . For an ego i , the constraint from an alter j is C_{ij} . The network constraint for an ego is the sum of the constraint index from all alters: $\sum_j C_{ij}$. Here,

$$C_{ij} = \left(p_{ij} + \sum_{q=1}^N p_{iq} p_{qj} \right)^2, \quad q \neq i, j$$

where $p_{ij} = z_{ij} / \sum_{q=1}^N z_{iq}$, z_{ij} refers to the intensity of the relationship from i to j , and is based on the relationship closeness question. As was mentioned above, participants reported their perception of each relationship between two alters as “close” or “positive, but not especially close”, coded as 2 and 1 respectively. Relationships not described in either of these ways were coded as 0. For each alter, participants also answered “how close do you feel to this person,” ranging from “very close,” “close,” “not so close,” and “distant.” On average, the participants listed 30% of their alters as very close, 42% as close, 24% as not so close, and 4% as distant. To be consistent with the tie intensity measure among alters, we coded “very close” and “close” as 2, and “not so close” and “distant” as 1. We calculated two network constraint measures to test our hypotheses. The first was *network constraint within organization*, where j and q were both within the ego's organization, and this constraint measure captures structural holes within the organization. The second was *network constraint across organizational boundary*, when j was inside the organization and q was outside, or vice versa.

Control variables

Network constraint outside the organization

It is important to control for the network constraint from the ties to alters outside the organization. Structural holes across organizational boundaries may be less valuable if ego's network constraint outside the organization is high, meaning the extra-organizational ties are more densely connected. Also, structural holes outside of the organization may reflect the respondent's preferences in terms of socializing with friends who tend to know or

not know each other. It can thus help to test the idea that an unobserved individual character trait might influence both structural holes and job performance.

Higher ranked ties

Participants indicated in the network survey whether each alter was of higher rank, the same rank, or lower rank than themselves. Previous work by Cross and Cummings (2004) has shown that ties with higher ranking colleagues tend to provide resources that are more valuable in facilitating performance at work. We counted the number of higher rank contacts in an ego's network. On average, half of the alters listed were described as higher in rank, and among the higher rank alters, 24.4% were outside of the organization. Within the organization, rank obviously refers to hierarchy. Outside the organization, the concept is generalized. For example, a salesperson might view a top executive at a customer organization to be of higher rank.

Ties inside and outside the organization

We controlled as well for the number of alters inside and outside the organization. We controlled for network size using these two variables, because past results have shown that these two types of contacts have differential effects on performance at work (Cross & Cummings, 2004).

Ego's background

We controlled for the participants' sex, race and age. Caucasian men were the comparison categories, and dummy variables indicated women and non-whites. Because the participants were Executive-MBA students, they came from different industries. Due to almost half of the participants coming from finance and banking, we constructed a dummy variable to indicate participants' industry background as finance/banking. We also controlled for participants' organizations' business segmentation (i.e., whether an organization operates in a single or multiple industries). About 40% of the organizational data was collected from the Compustat database and the rest through internet search.

Familiarity

The performance evaluators were asked to indicate how well they knew the target using a four-point scale ranging from “not well at all” (1) to “extremely well” (4). The average rating for familiarity was 3.02 ($SD = 0.36$). We were concerned that respondents embedded in cohesive networks might be more likely to nominate colleagues who are their friends in the workplace as evaluators and thereby garner biased evaluations. We therefore controlled for the average familiarity score of the performance raters.

Results

Table 4 shows descriptive statistics and correlations among the variables. Overall, the table shows significant correlations between the dependent and independent variables. The four performance measures were all positively associated at a moderate level ($.09 < r_s < .50$), which suggests that they are correlated but not identical. This is consistent with the results of the factor analysis, which indicate that they capture four distinct performance domains. The network constraint scores also display sensible associations. Network constraint across the organizational boundary was moderately associated with network constraint within the organization ($r < .24$, $p < .001$), but was not significantly associated with the number of contacts either inside or outside the organization ($p_s > .3$). This suggests that the degree of structural holes bridged across the organizational boundary captures a unique feature of the network structure that is independent of the mere number of ties on either side of the boundary.

⁴ We also used in supplementary analyses Poldony and Baron's (1997) density based measure to capture the degree of structural holes, and it yielded results comparable to those we report below.

Table 4
Means, standard deviations, and correlations.^a

		Mean	SD	1	2	3	4	5	6	7	8
1	Female	0.28	0.45								
2	Non-White	0.45	0.50	.06							
3	Age	39.08	4.66	-.16**	-.05						
4	Finance	0.42	0.49	.01	.04	-.10					
5	Business segmentation	0.55	0.50	.08	.02	-.21***	.09				
6	No. of Different Gender	6.93	5.20	.69***	.02	-.10	-.05	.08			
7	No. of Different Race	6.72	6.60	.16***	.63	-.10	.03	.08	.18**		
8	No. of Contact Outside Organization	11.35	5.77	-.09	-.03	.12*	-.08	-.24***	-.03	-.11*	
9	No. of Contact Inside Organization	10.31	5.48	.15	.01	-.12*	.02	.25***	.24***	.21**	-.76***
10	How Many More in the Network	28.08	48.60	.09	-.01	.02	.02	.07	.14**	-.08	.23***
11	No. of Contact from Higher Rank	10.90	4.09	.04	-.02	.18**	.09	.12	.11*	.07	.06
12	Within Org. Network Constraint	0.37	0.21	-.09	-.11*	.20***	-.05	-.14*	-.16**	-.12*	.49***
13	Outside Org. Network Constraint	0.29	0.20	.09	-.01	-.08	.05	.16**	.01	.12*	-.54***
14	Cross Org. Network Constraint	0.16	0.17	-.02	-.12*	-.01	-.10	.04	-.04	-.08	.02
15	Rater Familiarity	3.02	0.36	-.08	.01	.07	.10	.05	-.10	-.05	.08
16	Creativity	5.92	0.43	.04	.09	.00	-.02	-.08	.04	.09	.09
17	Decision-Making	5.53	0.63	-.01	.05	-.08	.02	-.01	-.03	.08	.04
18	Task Execution	6.12	0.36	.13*	-.01	.03	.08	.08	.07	.06	-.01
19	Teamwork	4.69	0.36	.24***	-.02	-.01	.02	.04	.14*	.04	.04
10	How Many More in the Network	.01									
11	No. of Contact from Higher Rank	.23***	.15**								
12	Within Org. Network Constraint	-.58***	-.02	-.05							
13	Outside Org. Network Constraint	.41***	-.12*	.01	-.22***						
14	Cross Org. Network Constraint	-.05	-.01	.02	.24***	.01					
15	Rater Familiarity	-.10*	-.01	-.03	.08	-.03	-.06				
16	Creativity	-.08	.06	-.12*	-.02	-.02	-.19**	.25***			
17	Decision-Making	-.02	-.04	-.13*	-.03	-.10	-.16*	.04	.35***		
18	Task Execution	-.01	.01	-.04	.10	-.02	-.01	.19**	.50***	.34***	
19	Teamwork	-.01	.06	-.08	.13*	-.04	.05	.11*	.24**	.09	.34***

^a Two-tailed tests, * $p < .05$; ** $p < .01$; *** $p < .001$.

Preliminary analysis

Before testing our hypothesis, we examine the effect of the overall network constraint score on the overall job performance score, as well as the four performance dimensions respectively. Table 5 reports the regression results. We first tested the effect of the overall network constraint on the overall performance, represented by an average score of all four performance measures. A significantly negative coefficient term suggested that managers with a low overall network constraint had better overall job performance ($\beta = -0.47$, $t = -2.67$, $p < .008$). Then, we tested the overall network constraint on the four performance measures independently, and found only one significant effect on decision-making performance ($\beta = -1.38$, $t = -3.74$, $p < .001$). These results suggested that had we not decomposed the job performance and network constraint measures, we would have lost the divergent and distinct effects of with- and cross-boundary network constraints on different aspects of job performance.

Hypotheses testing

Table 6 reports the regression results. For each of the four dependent measures, we ran two OLS regressions.⁵ The first model for each dependent variable was a baseline model including only the control variables. The second added the variables testing our hypotheses. The results support each of the four hypotheses. The coefficient of the cross-boundary network constraint term indicates that managers who spanned more cross-boundary structural holes, indicated by a lower constraint score, showed better creativity performance ($\beta = -0.39$, $t = -2.73$, $p < .006$) and decision-making performance

($\beta = -0.57$, $t = -2.62$, $p < .009$). On the other hand, the network constraint within organization measure shows that managers with high network constraint at work, or the absence of structural holes within the organization, had better performance in terms of task execution ($\beta = 0.32$, $t = 2.55$, $p < .011$) and teamwork ($\beta = 0.31$, $t = 2.49$, $p < .013$).

Among the control variables, one interesting result was that managers with more higher-rank ties within the organization received lower evaluations for creativity ($\beta = -0.01$, $t = -1.85$, $p < .07$), decision-making ($\beta = -0.02$, $t = -2.00$, $p < .05$), and teamwork ($\beta = -0.01$, $t = -2.14$, $p < .03$), but not for task execution. This may be because high-ranking ties make an employee more committed to an organization's status quo and more accountable to prevailing norms. It may also be because cultivating higher-rank ties as a job-performance strategy is a poor fit with performances requiring divergent thinking, as these types of performance often flourish in the absence of hierarchy (Meltzer & Salter, 1962). Further, this negative influence of high-rank ties is compelling in light of the findings of Cross and Cummings (2004), which indicate that high-rank ties are associated with good overall job performance ratings, and it reinforces our position that it is important to decompose overall ratings into subcomponents that reflect different aspects of a job.

In addition, we also conducted an analysis to test the effect of cross-boundary structural holes and within-organization structural holes on the overall job performance scores. The result suggested that on average the cross-boundary network constraint had a negative effect ($\beta = -0.22$, $t = -2.09$, $p < .038$), whereas the network constraint within organization measure showed a marginally significant effect ($\beta = 0.18$, $t = 1.65$, $p < .10$). The less clear pattern of the within organization network constraint measure on overall job performance suggests that without differentiating job performance into four different domains, we may fail to appreciate the benefits of closing structural holes within an organization.

⁵ We also conducted additional analyses to test the effect of internal network constraint and the cross-boundary network constraint in separate models. The separate models lead to almost identical results as the models testing the effect of these two types of constraints simultaneously.

Table 5Results of the regression analyses for the overall network constraint scores.^{a,b}

	Overall performance	Creativity	Decision-making	Task execution	Teamwork
Constant	5.34 (0.264)	5.445 (0.361)	6.201 (0.547)	5.626 (0.314)	4.146 (0.31)
Female	0.129* (0.051)	0.027 (0.071)	0.074 (0.106)	0.139* (0.061)	0.273*** (0.061)
Non-white	−0.027 (0.044)	0.013 (0.06)	−0.024 (0.09)	−0.070 (0.052)	−0.035 (0.051)
Age	0.001 (0.004)	0.001 (0.005)	−0.007 (0.008)	0.005 (0.005)	0.005 (0.005)
Finance	0.025 (0.035)	−0.008 (0.048)	0.048 (0.072)	0.041 (0.041)	0.018 (0.041)
Business segmentation	−0.02 (0.035)	−0.092* (0.048)	0.028 (0.073)	−0.015 (0.042)	0.006 (0.041)
Number of different gender	−0.005 (0.005)	0.002 (0.006)	−0.013 (0.01)	−0.003 (0.006)	−0.007 (0.006)
Number of different race	0.006‡ (0.003)	0.007 (0.005)	0.009‡ (0.007)	0.007 (0.004)	0.003 (0.004)
Network size	0.002 (0.007)	−0.003 (0.009)	0.003 (0.014)	−0.004 (0.008)	0.009 (0.008)
Number of contacts from higher rank	−0.014** (0.005)	−0.014* (0.006)	−0.025* (0.01)	−0.006 (0.006)	−0.012* (0.006)
How many more in the network	0.001 (0.001)	0.001 (0.001)	−0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
Rater familiarity	0.145** (0.047)	0.253*** (0.064)	0.093 (0.097)	0.151* (0.056)	0.083* (0.055)
Overall network constraint	−0.474** (0.189)	−0.340 (0.257)	−1.384*** (0.392)	−0.171 (0.224)	−0.014 (0.221)
R-square	0.11	0.1	0.09	0.06	0.10
F	2.99	2.71	2.42	1.68	2.93

^a Beta with standardized error in parentheses.^b Two-tailed tests, ‡ $p < .1$, * $p < .05$; ** $p < .01$; *** $p < .001$.

Discussion

This study elucidates the theoretical link between network structure and job performance. Our model builds on the idea that both structural holes and network closure in a network provide social capital (e.g., Adler & Kwon, 2002; Baker & Obstfeld, 1999; Burt, 2005) but adds significantly to past examinations of this idea, through a decomposition of job performance that allows more direct documentation of the mechanisms through which structural holes and closure operate. We also offer a new theoretical argument that structural holes and network closure ideally occur in different parts of the network. When intra-organizational networks matter, it is better if they are less sparse and more cohesive. When organizational-boundary-spanning ties matter, it is better that they contain structural holes. The results show that spanning structural holes across the organizational boundary is a better predictor of creativity and decision-making performance than spanning holes either within or outside the organization. Alternatively, for execution and teamwork, spanning fewer structural holes inside the organization outweighs the benefits of a sparse network. Our finding suggests that an optimal network structure for achieving higher job performance is to create structural holes across the organizational boundary and to close structural holes within the organization.

The results also speak to the classic concept of *tertius gaudens* (Simmel, 1950) by looking at the interaction of two dimensions of a basic triad (an ego and two alters), whether there would otherwise be a structural hole between the two alters, and whether the two alters are inside or outside the organization (see also Gould & Fernandez, 1989). The information advantage Burt (1992) associated with structural holes roughly corresponds to the advantageous position occupied by the gatekeepers in the earlier innovation literature to the extent that both imply boundary positions (Allen, 1977). Gatekeepers serve as the primary link to external sources of information and technology and facilitate information flow across the organizational boundary (Tushman,

1977). In this tradition, the value of the external information to the organization and to the ego is often treated as identical. However, in the context of job performance evaluation, the value of being a gatekeeper for the ego drops dramatically if s/he is not uniquely connected to the external contact. Likewise, the benefit of spanning structural holes within the organization is limited, because ego is not the unique person who can provide valuable information to the organization. Our results therefore amplify previous findings that the value of structural holes depends on the competition dynamics around the ego (Burt, 1997, 2005). In this regard, this paper enriches our understanding of when and how structural holes become competitive advantages.

By identifying that the location of the structural hole is a contingency variable, our finding also sheds light on the condition in which creativity performance is more likely to be associated with spanning structural holes. It is important to differentiate intra-organizational networks from extra-organizational ties (Rodan & Galunic, 2004). Structural holes within the organization may be a proxy for boundary-crossing holes—in our study intra-organizational and boundary-crossing network constraint scores were correlated. The possibility of confounding the effect of intra-organizational density by omitting boundary-crossing ties is further suggested by the fact that past work has found inconsistent relationships between intra-organizational density and creativity (Obstfeld, 2005). It may also be that in some very large organizations (e.g., global corporations), or organizations with high internal heterogeneity of knowledge among departments and employees, ties within the organization may be sufficiently separated from each other as to essentially be the equivalent of extra-organizational ties in terms of supplying novel information. Our results suggest that future research should examine the conditions under which organizational structure and demographics make intra-organizational structural holes more or less relevant for creativity performance.

These findings have significant practical implications. Managers are liaisons and information gatekeepers (Kotter, 1982; Luthans, Rosenkrantz, & Hennessey, 1985; Mintzberg, 1973). They make

Table 6
Results of the regression analyses for hypotheses testing.^{a,b}

	Creativity		Decision-making		Task execution		Teamwork		Overall performance	
Constant	5.219*** (0.321)	5.364*** (0.323)	5.555*** (0.492)	5.765*** (0.497)	5.590*** (0.279)	5.509*** (0.280)	4.239*** (0.275)	4.121*** (0.277)	5.139*** (0.236)	5.174*** (0.239)
Female	0.015 (0.071)	0.013 (0.070)	0.055 (0.109)	0.050 (0.108)	0.136** (0.062)	0.129** (0.061)	0.275*** (0.061)	0.268*** (0.060)	0.121** (0.052)	0.116** (0.052)
Non-white	0.018 (0.060)	-0.000 (0.060)	-0.017 (0.092)	-0.041 (0.093)	-0.071 (0.052)	-0.052 (0.052)	-0.042 (0.051)	-0.020 (0.052)	-0.026 (0.044)	-0.026 (0.045)
Age	-0.000 (0.005)	-0.001 (0.005)	-0.009 (0.008)	-0.010 (0.008)	0.005 (0.005)	0.003 (0.005)	0.004 (0.005)	0.003 (0.005)	-0.000 (0.004)	-0.001 (0.004)
Finance	-0.009 (0.048)	-0.025 (0.048)	0.043 (0.074)	0.020 (0.073)	0.038 (0.041)	0.042 (0.041)	0.018 (0.041)	0.026 (0.041)	0.023 (0.035)	0.018 (0.035)
Business segmentation	-0.088* (0.048)	-0.078 (0.048)	0.039 (0.074)	0.053 (0.074)	-0.015 (0.042)	-0.011 (0.042)	0.008 (0.041)	0.009 (0.041)	-0.016 (0.036)	-0.009 (0.036)
Number of different gender	0.003 (0.007)	0.003 (0.006)	-0.011 (0.010)	-0.011 (0.010)	-0.002 (0.006)	-0.002 (0.006)	-0.007 (0.006)	-0.006 (0.006)	-0.005 (0.005)	-0.004 (0.005)
Number of different race	0.007 (0.005)	0.007 (0.005)	0.009 (0.007)	0.010 (0.007)	0.007* (0.004)	0.006 (0.004)	0.003 (0.004)	0.002 (0.004)	0.007* (0.003)	0.006* (0.003)
Number of contacts inside the organization	0.001 (0.008)	-0.002 (0.009)	0.022* (0.013)	0.019 (0.013)	-0.003 (0.007)	0.004 (0.007)	0.006 (0.007)	0.012* (0.007)	0.007 (0.006)	0.009 (0.006)
Number of contacts outside the organization	0.005 (0.008)	0.004 (0.008)	0.021* (0.012)	0.018 (0.012)	-0.004 (0.007)	-0.005 (0.007)	0.007 (0.007)	0.006 (0.007)	0.007 (0.006)	0.006 (0.006)
Number of contacts from higher rank	-0.013** (0.007)	-0.012* (0.007)	-0.022** (0.010)	-0.020* (0.010)	-0.006 (0.006)	-0.006 (0.006)	-0.011** (0.006)	-0.012* (0.006)	-0.013* (0.005)	-0.013* (0.005)
How many more in the network	0.001 (0.001)	0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Rater familiarity	0.245*** (0.064)	0.236*** (0.064)	0.069 (0.098)	0.053 (0.098)	0.150*** (0.056)	0.141** (0.056)	0.082 (0.055)	0.076 (0.055)	0.136*** (0.047)	0.126*** (0.047)
Outside organization network constraint	0.050 (0.136)	0.049 (0.135)	-0.398* (0.208)	-0.397* (0.206)	-0.105 (0.118)	-0.098 (0.117)	-0.100 (0.116)	-0.094 (0.115)	-0.141 (0.100)	-0.137 (0.099)
Cross-boundary network constraint		-0.386** (0.140)		-0.573** (0.217)		-0.018 (0.122)		0.094 (0.120)		-0.217* (0.104)
Within organization network constraint		-0.020 (0.143)		0.040 (0.221)		0.322* (0.124)		0.309* (0.122)		0.175† (0.106)
F	2.39	2.67	1.43	1.74	1.58	1.86	2.73	2.98	2.32	2.42
R-squared	.093	.117	.058	.080	.063	.084	.104	.129	.091	.108

^a Beta with standardized error in parentheses.

^b Two-tailed tests, † $p < .1$, * $p < .05$, ** $p < .01$, *** $p < .001$.

and maintain contacts and interactions outside the formal organizational chain of command; they monitor and disseminate information among co-workers, subordinates, as well as superiors. Essentially, these actions involve establishing and developing various social network ties. While much evidence has suggested that managers spend a significant amount of time at work networking, we know less about where they build what kinds of network structures leading to which performance outcomes (Luthans, Hodgetts, & Rosenkrantz, 1988; Luthans et al., 1985). Our results provide nuanced advice in terms of where to build what kinds of network for what types of performance. With this knowledge, managers could also better prioritize resource and effort in either connecting colleagues at work or building new contacts outside the workplace depending on which performance dimension is most relevant to their overall performance and long-term career goals.

Limitations and future research

This study examined only egocentric networks. The strength of the egocentric network is that it is not constrained by a network boundary imposed by researchers, so this study capitalized on the broader scope of egocentric networks and examined network structures both within and across organizations' boundaries. But a key trade-off is that we were not able to consider structural characteristics of the networks beyond one node from the survey respondent. In addition, perceptions of networks may be partially driven by the organization in which a manager is embedded, although our results were robust to industry controls and to a rough measure of organizational size that we used as a control variable in supplementary analyses. At the same time, the fact that our

participants came from more than 100 organizations has its own advantages in terms of generalizability.

A second limitation stemming from the ego-centric survey method was our inability to validate the existence of the reported relationships between alters (Krackhardt, 1990; Krackhardt & Kilduff, 1999). However, previous research suggests that people are able to accurately report typical social relationships (Hansen, 1999; Marsden, 1990). It seems unlikely that people who are good at certain types of performance, such as creativity or decision-making, would systematically recall their network in a biased fashion that results in differential accuracy inside and outside the organization. Such threats to reliability and validity therefore are not of great concern in this study.

Furthermore, our survey asked the managers to report contacts whom they thought of as important in their networks. This may have biased them toward reporting contacts with whom they had positive relationships. If so, we may have missed examining hindrance ties that would also be expected to impact performance. Negative relationships, in which one person indicates preferring to avoid another, are known to be related to higher perceived conflict at work (Labianca, Brass, & Gray, 1998). Sparrowe et al. (2001) found that individual job performance was negatively related to centrality in hindrance networks, where they defined hindrance as a link that creates difficulty in carrying out job responsibilities. Future research might use an alternative name generating method that can more effectively elicit both positive and negative ties.

Prior research on boundary spanning has shown that ties crossing departmental or functional boundaries are effective for promoting knowledge transfer within an organization (Cross &

Cummings, 2004; Hansen, 1999; Szulanski, 1996; Tsai, 2001). Interestingly, in this study we did not observe any difference in the influence of intra- and inter-unit ties. The benefits of inter-unit ties within an organization may be particularly important in knowledge-intensive work, but our data included managers from a variety of organizations, where differences among industries and organizations may have prevented us from identifying any inter-unit benefits. Yet, because our respondents did many jobs in many companies, we are in a better position to claim that the relationships between network structure and job performance we found are general, and not dependent on the culture, formal structure, or the nature of jobs within any one organization.

Our model opens a number of promising directions for future studies. First is to examine whether the tie content would interact with the tie location (e.g., within vs. outside the organization boundary) in predicting performance at work. Specifying the tie contents would allow us to take a deeper look at the social and political influence of network on job performance. The ideal structure for various job performance measures may depend on the content of each tie. For example, Podolny and Baron (1997) showed that an open information advice network and a closed mentorship network are critical for career development. Morrison (2002) found that having a high density informational network increased both task mastery and role clarity among newcomers at work, while density of the friendship network had no impact. And most relevant to our study, Gargiulo et al. (2009) found that investment bankers received higher bonuses if their network of internal clients was sparse and their network of internal supporters was dense. Their explanation was that performance for bankers depends sometimes on being the source of unique ideas and sometimes on getting things done, and that different networks produce those outcomes. Our multidimensional examination of performance gives credence to their explanation by showing that indeed brokerage and closure matter differentially for different dimensions of performance. Likewise, we think that studies that consider the content of different types of networks would also benefit from the practice we introduced here of differentiating network structure according to the organizational boundary.

Whereas one contribution of the current paper is to identify the performance domains that pertain to a person's social network at work, this is only an initial step in understanding how people could draw on social capital to achieve better performance. Our results show a promising direction for the future: studying the effect of social networks on more nuanced performance domains, such as motivating others at work or enhancing conflict resolution performance. In addition, underlying our research is the assumption that social networks affect actual performance outcomes and thereby affect performance evaluation ratings. Alternatively, social networks may also affect the perception of a person's performance at work, without changing the actual job performance per se.

Given the significance of network structure in predicting performance, another question for future research is about the social networking orientation that can give rise to effective network structures. For example, to predict a manager's ability to implement innovations, Obstfeld (2005) found that managers with high *tertius iungens* orientation, those who have a proclivity for introducing two previously unrelated contacts, have better innovation performance. Obstfeld's innovation involvement measure captures contribution implemented changes in a comprehensive product, which can thus be understood as instances of execution and teamwork. In this way, the advantages he identifies of *tertius iungens* are in line with those we find for inter-organizational constraint. More recently, Baer (2010) has shown that even if managers are highly motivated to implement their creative ideas, they are more likely to succeed if they are high on a measure that captures

general social networking ability (Ferris et al., 2005). Such empirical findings lead to the broader theoretical question as to how and why social networking contributes to job performance. Future studies should seek to identify various networking strategies, inside or outside the workplace, which could lead to distinct network structures and ultimately affect a variety of types of performance at work. The search for those strategies will benefit substantially from the results presented here, which add precision to the understanding of what network structures are associated with what types of performance.

In conclusion, this study enriches our understanding of the link between network structures and job performance in two ways. First, by examining network structure and boundary spanning simultaneously, this study has provided a coherent and useful theoretical account of the performance implications of closure and brokerage in networks at work. Second, the relationship between network structure and job performance is contingent on the nature of the performance dimension. These two advances combine to produce a much clearer understanding of what structure, in what part of a professional network, influences what dimensions of job performance. Brokerage across organizational boundaries produces creativity and decision-making benefits, while closure within the organization is associated with improved task execution and teamwork.

Appendix A

Creativity:

1. S/he is able to work to implement new ideas.
2. S/he is good at coming up with creative ideas.
3. S/he is able to find improved ways to do things.

Decision making:

1. S/he rushes to judgment, rather than carefully weighing the consequences of various possible actions. (R)
2. When making a decision, s/he seeks information from a diverse array of resources.
3. S/he defines problem effectively and gets to the heart of a problem.

Task execution:

1. S/he can be relied onto complete a task s/he has agreed to do for me.
2. S/he maintains focus on the task at hand.
3. S/he can be relied onto have the knowledge and competence for getting tasks done.

Teamwork:

1. When working on a group project, s/he tends to want to do it all him/herself. (R)
2. S/he takes initiatives in contributing to the team's efforts.
3. S/he is unwilling to sacrifice his/herself interest for the good of the team. (R)

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